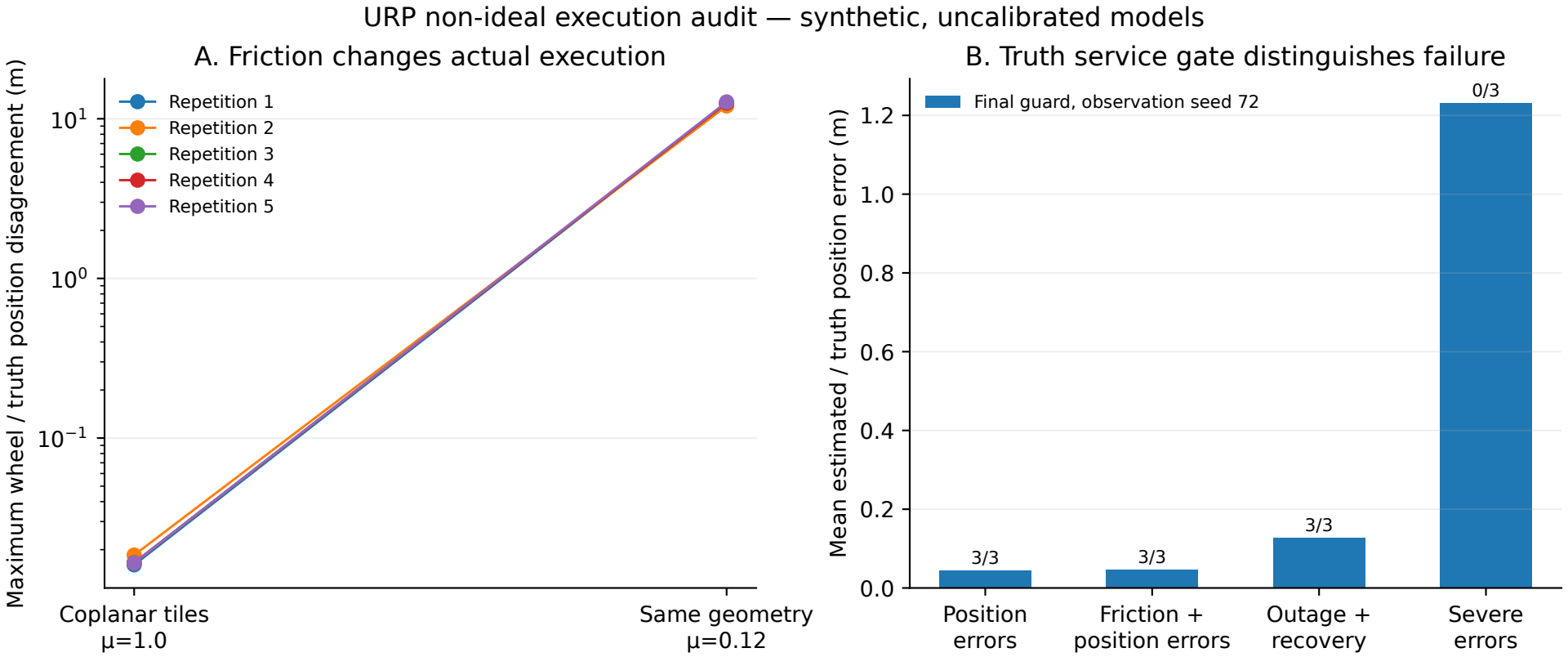
URP non-ideal execution audit — synthetic, uncalibrated models
A. Friction changes actual execution
Maximum wheel / truth position disagreement (m)
Repetition 1
Repetition 2
Repetition 3
Repetition 4
Repetition 5
10^{1}
10^{0}
10^{-1}
Coplanar tiles
μ=1.0
Same geometry
μ=0.12
B. Truth service gate distinguishes failure
Mean estimated / truth position error (m)
Final guard, observation seed 72
0/3
3/3
3/3
3/3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Position errors
Friction + position errors
Outage + recovery
Severe errors
A: five sessions per condition (truth control); B: final guard, labels are truth-verified tasks.
Same frozen route within each run; no population confidence or agricultural calibration claim.